Roll Transparency Log: Initiation document · Case 03

Following chapter 1 (Initiation) of the Magshimim project-book template: description, goal, audience, existing solutions, technological core, technologies, challenges and sources, hardware, conclusion.

Description

A system that records every change to the electoral roll (addition, deletion, correction) in a public, append-only log in which no entry can be silently removed or altered. Each entry carries a specific reason code ("shifted residence", "deceased", "duplicate"), a date, a polling booth and a pseudonymous id of the approving officer.

The log is a Merkle tree, exactly like the logs that govern web security certificates today (Certificate Transparency). Every few minutes it publishes a signed tree head: one value that summarises the whole history. Any voter can check, on a phone, by SMS or at a booth kiosk, what happened to their own entry and receive a short mathematical proof that it is really in the log. Independent monitors (parties, journalists, civil-society groups) check that today's tree head extends yesterday's, so nobody rewrote the past.

Goal

The core problem in the story is not deletion as such. A revision was overdue and many deletions are correct. The problem is that nobody, including the person removed, can see what happened to their name, why, and until when it can be fixed. What cannot be seen cannot be checked, and every mistake looks like intent.

Goal: turn every change to the roll into a visible, verifiable event without exposing personal data, so that the truth about the roll does not depend on trusting whoever runs it.

Audience

- **Primary:** the Election Commission (ECI) and district electoral registration officers, who run the roll and need to show it is clean.
- **Wide:** every voter, through a portal, SMS or kiosk. Required knowledge: none. The output is one sentence: "You were removed on 14 Nov, code R-SHIFT, 9 days left to appeal".
- **Monitors:** parties, journalists, researchers and NGOs running monitors that check the log was never rewritten.
- **Scenario:** Priya, a nurse in Lucknow, gets an SMS on the day her entry in her parents' village is removed. She does not wait for the draft, search by herself or guess which of three reasons applies.

Existing solutions

- **Supreme Court of India order (August 2025):** in the Bihar revision the Court directed the ECI to publish the ~65 lakh removed names, booth-wise, with reasons. Transparency is legally and practically possible, but a static list neither proves it was not changed later nor notifies the person.
- **ECI electoral search portal:** searches the current roll. It shows state, not history: if you were removed, you are simply not there.
- **Certificate Transparency (RFC 6962, RFC 9162):** since 2013 every web certificate is logged in public Merkle logs checked by monitors at the scale of billions.
- **Trillian, Sigstore Rekor, Key Transparency:** open-source transparency logs for software and keys. Building blocks, not a roll solution.

What is new here: a transparency log on the roll itself, with mandatory reason codes, proactive notification and a privacy layer that lets everyone check themselves without exposing others.

Technological core

An append-only Merkle tree. Two short proofs carry all the value: an **inclusion proof** ("your entry is in this tree") and a **consistency proof** ("today's tree contains yesterday's unchanged"). Both are logarithmic: about 27 hashes for 130 million entries. Without this core there is no project.

Main technologies

- **Trillian (Go):** Google's open-source log engine for certificate transparency, proven at billions of entries.
- **SHA-256 and Ed25519:** hashing and tree-head signatures.
- **HMAC with a secret pepper in an HSM:** a stable voter pseudonym that cannot be reversed to the elector ID.
- **PostgreSQL and Kafka:** intake of officer decisions before they enter the log.
- **Android/Web and an SMS gateway:** the voter interface.

Open questions: a public blockchain versus a Merkle log with independent witnesses (the latter gives the same no-silent-rewrite guarantee at negligible cost); and how much detail each public entry may carry. That is a question for the DPIA.

Challenges and sources

- **Privacy:** a log of who moved where is also a map. Irreversible pseudonyms, coarse codes and delayed publication for small booths.
- **Who feeds the log:** the log proves that what was written was not changed, not that it was right. It must be wired to the decision point (the officer terminal) so that no roll change exists without a log entry.
- **Voters without smartphones:** SMS, kiosks and booth-level officers as intermediaries.
- **Adoption:** the ECI must want it; the 2025 court precedent helps.

Sources: RFC 6962, RFC 9162, Trillian documentation, certificate.transparency.dev, reports on the Supreme Court's Bihar order (August 2025).

Hardware

None for the PoC: a laptop builds a one-million-entry log in under 20 seconds. In deployment: log servers, an HSM for signatures and the pseudonym key, and three to five independent monitors.

Conclusion

✅ Survives. The core is proven at a larger scale than any electoral roll and there is legal precedent for publishing removals with reasons. Condition: privacy. Research must show the log is useful to the voter and safe for the neighbours.